

AI Image Detection & Digital Forensics

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Abstract: The constant advances in generative artificial intelligence, specifically of models like Generative Adversarial Networks (GANs) and Diffusion models have evolved into generating photorealistic images. This drastic development has also raised some concerns related to the use of AI image generation for spreading misinformation, creating deepfakes, which leads to decreasing trust in digital media. As these concerns grow rapidly, a need for AI image detectors also increases and while the current detection methods have attained high accuracies on standard academic benchmarks and metrics, it still fails to deliver high accuracy results when it comes to real-world scenarios leading to severe brittleness. These detectors suffer from poor generalization when introduced with unseen generative architectures, multiple semantic content, and common image transformations. This paper explores these inconsistencies using a four-category taxonomy, consisting artifact-based, spectrum-based, image-encoder-based, and generator-aware methods, to inspect their robustness and generalization challenges. Through experimentation, testing and evidence it is known that shortcut learning has a major role in failure of detection. Here, detectors extensively overfit to biased dataset semantics and preprocessing artifacts, instead of learning on universal forensic traces of synthetic generation. To overcome this generalization challenge a fundamental reform of artifact-preserving training model, expansion of evaluation benchmarks, and implementation of real-world deployment strategies is needed in action.

1. Introduction

Generative AI has rapidly moved from research labs into daily and professional tasks, one of the domains that it has been dominating in is image generation. In recent years, there has been a significant improvement in AI-generated images, from generating non-uniform, broken images to now creating realistic images with detailed textures, lighting, styles and looking almost identical to a real life taken image. Visible artifacts have decreased with each new model generation. These improvements are

worthwhile to be noticed but the unethical use of AI image generation can't be overlooked and ignored. In recent times, there have been many such incidents where the use of image generation has caused issues like, during early 2026, a group of users on X (formerly known as Twitter) started using Grok to harass public images of women and underaged girls [22]. Such misuse is not isolated. There have been many such events like that where deepfakes, AI generated images or videos are revolving around the

internet to bring someone down politically or personally. As the AI generated images get better, a well made AI-generated image detector becomes a necessity.

The foundation of modern AI image generation is the generative architecture such as **Generative Adversarial Networks (GANs)** and **Diffusion Models**. GANs operate using two competing neural networks [1], which includes a generator that creates synthetic images and a discriminator evaluating their authenticity enabling rapid image generation. On the other hand, diffusion models are different, as they start from a random noise and gradually refine [2], resulting in a highly detailed and realistic structured image. Diffusion models have become much more mainstream than GANs especially for complex image synthesis tasks in the recent image generation models like DALL-E3, Google Gemini-based image models, ChatGPT Images etc. This architectural shift brings fundamental differences in artifacts left by GANs and diffusion models, which has significant implications for forensic detection.

Although generation has improved rapidly, detection has not kept pace. There are several challenges faced by AI-generated image detectors or models. In most cases, these models perform exceptionally well on the images they were trained on but they suffer from brittleness, when introduced to a new data or distribution shift or domain gap. Most of the existing research has focused on cross-generator generalization [8],[10], where detectors trained on one model family

(GANs) are evaluated against unseen models (like diffusion-based), detectors still fail majorly when encountering new image scenes or datasets with different semantics [9],[11], known as cross-scene or cross-domain generalization. The detectors might perform well on benchmarks, but when it comes to the real world where they encounter unseen semantic content which was entirely absent from their training set, the adaptability of detectors becomes restricted. This also creates another issue of transitioning across entirely different architectures like from GAN-based training data to Diffusion-based test data [7], which remains to be the primary constraint often requiring adaptive debiasing or specialized data augmentations for bridging the domain gap. The reason why this major gap and flaws exist is due to the trap of shortcut learning where they rely on different “shortcuts”[9]. Like semantic bias, where detectors overfit to a specific semantic bias present in the training set as the deep networks mainly focus on global semantics they often base their classification. Both real and generated images contain “semantic artifacts”[9] derived from their datasets and during training. Models overfit to these specific training-set artifacts which leads to significant accuracy drops when tested on new datasets. This leads us to a guaranteed failure in attaining a higher accuracy on standard benchmarks for reliable deployment. It’s also noticeable how most of the reported performance metrics exaggerates the accuracy of detectors which is due to the fact that current benchmarks barely reflect real-world distribution shifts [9],[12]. It’s important to note that this problem isn’t

about the dataset not being large enough, which means increasing the training data won't solve this problem. This paper explores the failures of training and evaluation of these detectors. Re-examination is required for both detector training criteria and benchmarks used for evaluation to address these failures.

This review contributes: a structured taxonomy for current AI-generated image

detection methods, an analysis of the limitations of current evaluation and benchmarking practices, assessing robustness and generalization challenges across detection models, and identifying gaps with proposed directions for future research in real-world deployment.

2. Scope of the Review

This review examines AI-generated image detection, and detection methods applied to images produced by GANs and diffusion models. It is limited to image-level detection only; video-level deepfake detection is outside the scope of this paper. This review

focuses on machine learning-based forensic techniques rather than rule-based methods. A particular emphasis on identifying the limitations of current detection approaches and the emerging directions that address these limitations.

3. Background

3.1 AI Image Generation

Overview (GANs, Diffusion, why artifacts exist)

GANs were introduced by Goodfellow et al. in 2014 [1]. They proposed a generator-discriminator architecture where they trained two networks in a competitive, adversarial, “minimax” game to create synthetic data. The generator’s role is to map random noise to a synthetic image and then the discriminator differentiates between real and fake. Mainstream architectures like ProGAN, StyleGAN, and BigGAN [4]-[6] advanced convolutions, which produces periodic checkerboard artifacts [12] and abnormal local pixel correlations. Forensic

detection methods use such consistent, imperceptible, and abnormal artifacts as the primary signal to differentiate AI-generated content from real images.

Ho et al. in 2020 [2] introduced Diffusion models. Their formulation is known as Denoising Diffusion Probabilistic Models (DDPM) enabled high-quality image synthesis by iteratively reversing a noising process. While training, Gaussian noise corrupts an image progressively across multiple steps until it forms pure noise, then the model starts learning to reverse the process step by step in order to recover a clean image from random noise at inference time. Some modern variations like Stable Diffusion (Rombach et al., [3]) increase

efficiency by operating in a compressed latent space. This enabled the quick development of large-scale text-to-image models like DALL.E 2, Imagen, and Midjourney. Compared to GANs, these models leave detectable artifacts through convolutional up-sampling in the VAE decoder [12] and high-frequency suppression during denoising.

Generative Adversarial Networks (GANs) and diffusion models have entirely different mechanisms in generation and architecture construction, there is still some similarity between these techniques. In particular, each technique has unique, low-level statistical fingerprints within its output. This refers to checkerboard effects caused by GANs and high-frequency filtering found in diffusion models, digital fingerprints that enable the differentiation of generated data from authentic images. Modern forgery detection approaches do not depend on unusual semantic mistakes; instead, they make use of these low-level fingerprints to detect forgeries. Through focusing on such common characteristics, detection methods are able to effectively employ spatial and frequency domain features in image generation networks.

3.2 Digital Image Forensics Basics

Digital image forensics is described as a specific branch of science [21] aimed at analyzing digital images with respect to their authentication, origin identification, and integrity verification. Genuine natural images are produced by the combination of a number of operations including acquisition with

sensors, demosaicing and in-camera processing such as color correction and sharpening, resulting in certain statistically recognizable features. These features are important in forensic analysis, as the lack of such characteristics often speaks of a manipulated image or an artificially generated one. Specifically, artifacts related to image compression, especially JPEG, serve as strong clues because of their 8x8 block nature.

In digital forensics, images can be analyzed using two primary representations. These include the spatial domain which involves the analysis of raw pixel values and the frequency domain which consists of components obtained through Fourier or wavelet transformations. The frequency domain helps in identifying periodic distortions and high-frequency components caused by up-sampling processes. The real cameras used in capturing images produce noise patterns known as photo response non-uniformity (PRNU) patterns. This means that if the image was captured using any type of computer graphics software, it would lack such noise. Such noise plays an important role in forensic analysis as the image's presence, absence, and variations help to determine the authenticity of an image. Residual analysis is also commonly used to enhance these subtle traces [10] by detecting these noise components in the image.

3.3 Taxonomy of Detection Methods

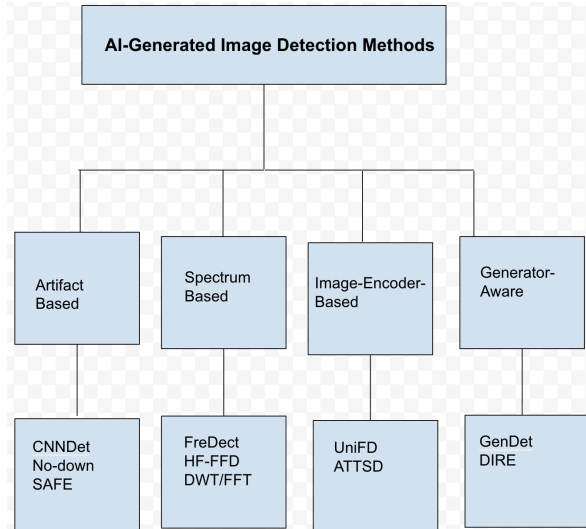


Figure 1. Classification of AI-generated image recognition techniques presented in this review based on the taxonomy proposed by Park et al. [7] but with an additional category of generator-aware techniques.

With time, AI-generated image detection methods have increased tremendously making it important to have categorization to understand their relative strengths and limitations. This paper adopts the three category taxonomy proposed by Park et al. [7], which are artifact-based, spectrum-based and image-encoder-based approaches, all of which depend on different image characteristics. The artifact-based approach is based on the detection of defects using neural networks, as seen in ResNet50 architecture. On the other hand, spectrum-based techniques use frequency fingerprints to detect abnormalities, which cannot be seen by the naked eye. The image-encoder-based technique uses pre-trained models like CLIP to apply generalization to recognize real and fake images easily.

Park's taxonomy relies on feature-based differences in detecting techniques, sorting detectors based on specific features utilized by each technique. In addition to the Park's three feature-based categories, there is another class worth discussing. There are detector methods utilizing information about generation rather than the extracted features only and these are referred as "generator-aware" methods. The difference between feature-based detectors and generator-aware detectors, besides performance, lies in viewing detection as a task of identifying anomalies or reconstruction. This approach of generating and detecting adversarial examples using the latter framework was outlined in GenDet (Zhu et al., 2023) where detection has been reframed as anomaly detection [13] with discrepancies between teacher and student being observed only on synthetic inputs but not on real inputs. The next several subsections discuss each detector type separately: 3.4 Artifact-based methods, 3.5 Spectrum-based methods, 3.6 Image-encoder-based methods, and 3.7 Generator-aware methods.

3.4 Artifact-Based / Spatial-Domain Methods

Artifact-based methods are specifically designed to operate directly on raw pixel values, training classifiers to detect inconsistencies left behind by the generation process. Such methods typically use convolutional neural networks (CNNs), where the variant ResNet is most commonly used to learn discriminative features that separate real images from the synthetic ones capturing fine-grained details like

“pixel-level inconsistencies” at the lower layers and broader structural patterns at higher layers. The type of artifacts that are targeted through this method are periodic checkerboard patterns from up-sampling, unnatural local pixel correlations, blending boundaries, and texture irregularities. The training paradigms utilize methods that are typically trained as binary classifiers on a dataset of real and synthetic images from one or a few specific generators. Some hybrid approaches combine multiple feature domains; for example, Yadav et al. [14] integrate multiscale CNN features with frequency-domain analysis and attention-based transformers within an ensemble framework for deepfake detection.

A foundational artifact-based baseline is CNNDetection (Wang et al., [15]), which uses a ResNet-50 binary classifier trained on ProGAN-generated images and utilizes data augmentation techniques like Gaussian blur and JPEG compression to improve its generalization capabilities. Significant research has refined this baseline approach, specifically the “No-down” detector deliberately removes downsampling [16] and pooling operations from the first ResNet layer to preserve subtle pixel-level artifacts that conventional pooling would otherwise suppress. Recently, Li et al. [12] introduced SAFE as a modern representative of this category, showing that simple shifts in the training paradigm matter more than complex feature engineering. These shifts include replacing resizing with cropping during preprocessing to prevent artifact distortion, and applying diverse augmentations like ColorJitter and a patch-based RandomMask,

SAFE achieves state-of-the-art generalization performance using only a lightweight network.

The main strength of artifact-based methods is that they are conceptually simple, computationally efficient, and can achieve exceptionally high detection accuracy on images produced by the specific generators they were trained on. However, a major limitation of this approach is their poor generalization to unseen generators, particularly across different architectural families, like the models trained exclusively on GAN-generated images often fail significantly [7],[8]] when evaluated on diffusion-generated images. Moreover, these methods are highly susceptible to shortcut learning, frequently overfitting to dataset-specific semantic characteristics [9]-[10] or localized training-set artifacts rather than acquiring universal forensic traces, a critical vulnerability that will be examined further in Section 5.

3.5 Spectrum-Based / Frequency-Domain Methods

Spectrum-based methods only analyze images in the frequency domain rather than invoking raw pixel values, Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), or Discrete Wavelet Transform (DWT) are methods used for transformation of images to their frequency components. The core logic for this approach is that generative models, regardless of their core architecture, inherently leave behind detectable signatures in the frequency domain; for example, GANs produce periodic peaks in the spectrum known as

checkerboard artifacts during the up-sampling operations, while diffusion models usually suppress high-frequency content and exhibit low-frequency energy concentrations, creating different but equally exploitable signatures. The main advantage of frequency over spatial analysis is that frequency-domain features are often more robust to image transformations like resizing or compression, and high-frequency components in particular generalize better across different unseen generators as most generative models keep struggling to faithfully reproduce intricate high-frequency details.

An early frequency-based detector, FreDect (Frank et al., [17]), utilized Discrete Cosine Transform (DCT) based features to identify GAN-specific spectral artifacts. Following this, the High-Frequency Face Forgery Detection (HF-FFD) approach (Luo et al., [18]), evaluated by Brodarič et al. [10], employs Spatial Rich Model (SRM) filters to extract high-frequency residuals as input to a classifier; critically, their Grad-CAM analysis revealed that methods with broader, more distributed high-frequency activation patterns generalize better than those focused on narrow, localized regions. Further complicating frequency analysis, Zheng et al. found that the frequency domain itself contains “semantic artifacts”[9], as even real images from different datasets exhibit distinct spectral fingerprints arising from their specific content and preprocessing. Their patch-shuffle approach forces detectors to learn local frequency features rather than overfitting to these dataset-level frequency patterns. Most recently, Li et al.’s SAFE

pipeline uses the Discrete Wavelet Transform (DWT) [12] to explicitly extract high-frequency (HH-band) components as input features, demonstrating that high-frequency information generalizes substantially better than low-frequency or full spatial features across both GAN and diffusion based generators.

The major strength of frequency domain methods is that it often generalizes better across architectural families because both GANs and diffusion models leave detectable spectral signatures, even when the spatial-level artifacts differ substantially. Despite the strength of frequency-based methods, they can be sensitive to image processing operations that affect the spectrum, such as resizing or strong compression, and may need careful preprocessing to preserve discriminative frequency patterns. A second limitation comes from Zheng et al. [9], who demonstrated that frequency-domain features are not immune to shortcut learning and semantic artifacts present in the spectra of training-set images can still cause overfitting. This partially undermines the common assumption that frequency features are inherently generator-agnostic.

3.6 Image-encoder-based Methods

Unlike any other methods, image-encoder-based methods exploit pretrained vision-language or vision-only models, most commonly CLIP, by extracting high-level semantic features from images and training a lightweight classifier on top of them. Through leveraging the rich representations that models like CLIP have

acquired through large-scale image-text pretraining, these methods bypass the need to learn low-level forensic features from scratch, effectively treating real-versus-synthetic image detection as a downstream classification task. An example of this approach would be of UniFD (Ojha et al., [19]) which utilizes the frozen feature space of CLIP:ViT-L/14 together with a simple linear classification layer to attain strong cross-generator generalization, particularly on images produced by diffusion models. Through this foundation, a recent advancement by Cai et al. [11] introduces Adaptive Test-Time Semantic Debiasing (ATTSD) extends the image-encoder methods by adding a self-supervised auxiliary task during testing that aligns visual-semantic through CLIP visual representation to address the problem of cross-semantic generalization. While image encoders show better generalization capabilities when compared to unknown generators, especially model diffusion models, compared to the conventional artifact-based models, they still tend to be less reliable on clean real-world images, inheriting the semantics learned by their pretrained vision-language models.

4. Datasets and Evaluation

Datasets play a vital and foundational role in AI-generated image detection, as they determine what generators a detector learns to recognize, what real images it detects as “natural”, and what failure modes go unnoticed during evaluation. A typical

3.7 Generator-aware Methods

Generator-aware techniques work on a different principle and make use of explicit information about the generation process rather than framing the detection task as a conventional binary classification problem on feature vectors. In most cases, generator-aware techniques frame the detection task as anomaly detection or image reconstruction; the premise here is that a generative model cannot perfectly reconstruct a real image, so reconstruction errors can serve as a forensic signal. For instance, Zhu et al.'s GenDet [13] leverages a teacher-student network setup that produces minor discrepancies between outputs when fed real images and significant discrepancies when dealing with synthetic images, with the discrepancy being used as a detection signal. Similarly, reconstruction-based techniques like DIRE [20] invert images through the diffusion process and measure the resulting reconstruction error as a detection signal. Generator-aware techniques show impressive generalization capabilities since they target the generation process rather than the artifacts specific to each dataset, but they are more computationally demanding compared to other techniques and rely on having information about the generation process.

detection dataset pairs synthetic images from one or more generators with real images sampled from a known source, common real-image sources include LSUN, ImageNet, COCO, and CelebA, and most datasets are released under permissive licenses such as CC0 or research-use terms. Detectors are

usually trained on images from a single generator (or a small set) and evaluated on held-out images from the same generator, then on images from unseen generators to assess their generalization. Table 1

summarizes the major datasets used across the works reviewed in this paper, along with their generator coverage, real-image sources, and notable characteristics.

Dataset	Year / Source	Generator Type(s)	Real Image Source	Approx. Size	Used By (in this review)	Notable Characteristic / Limitation
FaceForensics++ (FF++)	2019, Rossler et al.	GAN-based face manipulations (Deepfakes, Face2Face, FaceSwap, NeuralTextures)	1000 YouTube videos	1.7M+ frames	Brodarić et al. (2024), Song et al. (2023)	Face-only; primarily face-swapping; limited to GAN-era manipulations.
ForenSynths	2020, Wang et al.	11 generators (mostly GANs: ProGAN, StyleGAN, BigGAN, CycleGAN, etc.)	LSUN dataset	~720K images	Park et al. (2024), Li et al. (2025), Zhu et al. (2023)	Standard benchmark for GAN detection; underrepresents diffusion models.
GenImage	2024, Zhu et al.	8 generators (mostly diffusion: SDv1.4, SDv1.5, ADM, GLIDE, Wukong, VQDM, Midjourney + BigGAN)	ImageNet	~2.7M images	Park et al. (2024), Li et al. (2025), Zheng et al. (2024)	Largest diffusion-focused benchmark; varied image dimensions add real-world relevance.
DiffusionDB	2022, Wang et al.	Stable Diffusion v1.4	LAION-5B	~14M prompts	Zheng et al. (2024)	User-prompted images; reflects real-world generation behavior.
UniversalFakeDetect	2023, Ojha et al.	Extends ForenSynths with diffusion models (LDM, Glide, DALL·E)	LSUN	Built on ForenSynths	Zhu et al. (2023), Li et al. (2025)	Bridges GAN and diffusion eras; key benchmark for cross-architecture generalization.
DeepFakeFace (DFF)	2023, Song et al.	Diffusion-based (SD v1.5, SD Inpainting, InsightFace)	IMDB-WIKI (celebrities)	120,000 images (30K real / 90K fake)	Song et al. (2023)	Diffusion-era face dataset; tests cross-generator and degraded-image classification.

CSAID	2025, Cai et al.	Diffusion-based (SDv2, LDM, LCM)	COCO + PartiPrompts	12,000 images (1K real + 1K fake per category $\times$ 6 categories)	Cai et al. (2025)	First benchmark explicitly designed for cross-semantic generalization.
DiTFake	2025, Li et al.	DiT-based (Flux, PixArt, SD3)	COCO	~6,000 images (~3K per generator family)	Li et al. (2025)	Targets latest diffusion-transformer architectures (post-UNet era).
Kaggle 140K Real & Fake Faces	2020 (release), Kaggle	GAN-based (StyleGAN)	Mixed open-source	140,000 images (70K real / 70K fake)	Yadav et al. (2025)	Single-generator dataset; limited diversity; commonly used in entry-level deepfake studies.

Table 1. Major datasets used across the works reviewed in this paper, including generator coverage, real-image sources, and notable characteristics.

A persistent issue across the field is that detectors are trained on narrow, generator-specific datasets, causing them to recognize one generator’s signature rather than synthetic images in general, a phenomenon known as dataset bias. For example, the widely used ForenSynths dataset is dominated by GAN-based generators, while GenImage focuses on diffusion-based ones, so detectors trained on either dataset often fail to generalize to the other architectural family, as Park et al. (2024) and Song et al. (2023) demonstrate empirically [7]-[8]. Beyond this architectural bias, datasets also exhibit semantic bias: most commonly used datasets feature narrow content categories (faces in FF++, single-object scenes in ForenSynths), which Cai et al. (2025) argue causes detectors to overfit [11] to category-specific patterns rather than universal forensic features. This means that “high benchmark accuracy”

reported in the literature often reflects detection within the training dataset’s particular bias profile, not general detection capability.

Even after controlling for generator and semantic bias, detectors can still overfit to dataset-level artifacts arising from preprocessing, compression, or the specific image source itself. Zheng et al. (2024) demonstrated that real images from different datasets exhibit distinct frequency-domain “fingerprints” [9] stemming from preprocessing pipelines, a detector trained on one real-image source may inadvertently learn the source’s preprocessing artifacts rather than the synthetic-vs-real distinction. This issue is partially hidden by standard evaluation metrics: detectors often report high Average Precision (AP) on out-of-distribution data while showing severely degraded accuracy at standard

thresholds, indicating that ranking is preserved but decision boundaries collapse, a gap that benchmark numbers alone fail to expose [9]. Together, these issues suggest that current evaluation practices systematically underestimate the deployment gap of detectors trained on conventional datasets.

A growing trend in the field is to use existing generative models themselves to construct evaluation datasets, for example, Cai et al.’s CSAIID benchmark [11] uses Stable Diffusion v2, LDM, and LCM to generate semantically diverse images, while Li et al.’s DiTFake test set [12] uses Diffusion Transformer-based generators (Flux, PixArt, SD3) to evaluate detection on the latest architectures. This approach allows researchers to keep updated evaluation datasets with rapidly evolving generative models, but it also raises questions about whether evaluation generators are sampled diversely enough to reflect the full landscape of deployed models in real-world settings or whether they risk falling into dataset bias by catering to a specific subset of real-world data. The lack of standardized, regularly updated benchmarks remains an open challenge for the field.

5. Robustness and Generalization Challenges

In spite of all the methodological differences surveyed in Section 3, it has been observed that there is a common failure pattern among current detectors when faced with training distribution beyond the one they were trained on. This section covers four interconnected challenges in order to define the robustness

and generalization problems in AI-generated image detection. These challenges arise from a foundational issue where detectors learn dataset-specific shortcuts rather than universal forensic features.

5.1 Cross-Generator Generalization

Detectors trained on images created by one generator architecture consistently fail when tested on images created by another, especially when there is a higher architectural gap between the training and testing generators. For instance, Song et al. (2023) found that detectors trained on FaceForensics++ [8] (mostly GANs) could only detect diffusion-created faces in DeepFakeFace (DFF) with an accuracy of 38-58%, which is just above random chance levels for an unseen architectural family. This trend was validated by Park et al. (2024) on 23 different benchmark architectures [7]: detectors based on artifact extraction work poorly on diffusion-generated images compared to GANs, whereas image encoder-based models are poor at detecting GAN faces. Also, changing the generator from one architectural paradigm to another, such as going from convolutional diffusion (e.g., Stable Diffusion) to transformers (e.g., Flux, SD3), presents another generalization issue rarely addressed by current detectors. This is exemplified by Li et al. (2025) using their DiTFake benchmark [12], which evaluates models on diffusion images generated via transformers. This evidence suggests that the concept of “generator agnostic” detection is still an ideal, given that

most modern detectors can generalize within architectural paradigms but not across them.

5.2 Sensitivity to Image Transformations

In real-world deployment, images go through routine transformations such as JPEG, compression, resizing, blurring, and color shifts, long before they reach a detector, yet detector performance often decreases sharply under such conditions. Deepfake detectors were tested under common image perturbations like color saturation changes, contrast adjustments, Gaussian blur, and pixelation, Song et al. found inconsistent and at times counterintuitive results [8]. Gaussian blue, for example, improved detection accuracy by accident on images generated by Stable Diffusion while degrading on others. Similarly, Park et al. [7] observed that using a quality factor 50 for applying JPEG compression, reduced detection accuracy by nearly 30 percent points for some methods, mostly affecting the Vision

Transformer-based detectors that hadn't been trained with appropriate data augmentations. These observations tell us that robustness is largely determined by training-time augmentation choices rather than purely by detector's architecture, a point strengthened by Li et al.'s SAFE pipeline, demonstrating that combining simple augmentation approaches [12] like ColorJitter and RandomMask can significantly improve transformation robustness even when using a lightweight network. These vulnerabilities indicate that if robustness testing protocols aren't standardized, reported benchmark performance remains as a poor proxy for how

detectors will perform on the degraded images encountered in practical applications.

5.3 Real-World Deployment Gap

Deployment gap happens when both cross-generator brittleness and sensitivity to image transformations exist, which happens due to the existing differences between performance on the benchmark and actual ability to detect. The reality is that images posted on social networking platforms undergo several cycles of compression, resizing, color space transformations, and even filters, all while being generated from a far more diverse set of generators than any academic benchmark can hope to capture. Concrete evidence of the difference between deployment behavior and benchmark behavior was demonstrated by Li et al. [12], who indicated that when their SAFE detector was deployed on a production environment handling millions of images daily, the number of true positives detected was almost 50% greater at a similar level of precision than that achieved by the prior baseline detector that used FFT features. Bridging this deployment gap requires rethinking both evaluation protocols and detector design priorities.

5.4 Overfitting to Artifacts (Shortcut Learning)

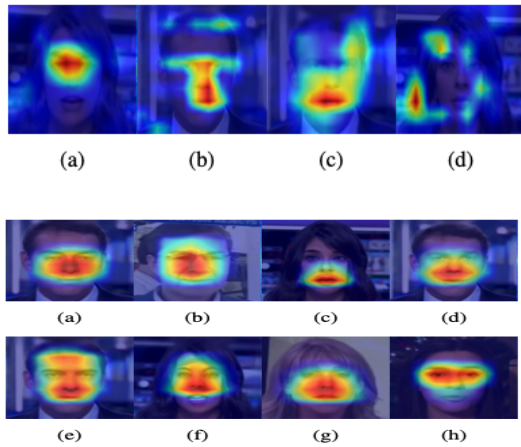


Figure 2. Grad-CAM visualization showing narrow, localized activations in poorly generalizing detectors versus broader, distributed activations in better-generalizing detectors. Adapted from Brodarič et al. [10].

In each of the above-listed difficulties, a similar cause is involved, namely, the tendency of detectors to learn artifacts instead of true indicators of fake images generated by synthesis methods. The direct proof of this process was found by Brodarič et al. (2024) by conducting the analysis of detector activations with Grad-CAM [10] visualization, showing that detectors demonstrating weak generalization capabilities on new datasets relied on a few localized regions, reflecting artifact patterns of the training generator, while more capable detectors featured more diffuse class activations. Also, Zheng et al. (2024) showed that the learned shortcut is not limited to the artifact patterns [9] of training generators; namely, detectors learn certain "semantic artifacts" defined as a distinctive fingerprint in the frequency domain, which exists only

because an image belongs to a particular dataset. Additionally, detectors trained on narrow semantics, for example, individual classes including only cars or cats, exhibit inferior performance [11] when tested on rich semantic images containing several different objects. This explains that current detectors are learning semantic properties of training data instead of true generation traces.

Lie et al. [12] reframed the problem one step ahead, by asserting that shortcut learning isn't just a failure from feature selection but it's also a training paradigm failure: most of the common preprocessing steps like resizing image actively distorts the very low-level traces that differentiate generated images from the real ones, while monotonous augmentation strategies fail to reveal diversity needed for real generalization to detectors. The above four lines of evidence points to the same diagnosis again that the major failure of current detectors isn't insufficient model capacity, insufficient training data, or a weak architectural design, but a system level issue which exploits dataset-specific shortcuts that standard benchmark evaluations fail to detect. It requires thorough re-examination of both the training paradigms through artifacts-preserving, preprocessing, diverse augmentation, and patch-based local feature learning and evaluation protocols, which should be reimagined to expose reliance on shortcut instead of rewarding it.

6. Discussion

The issues discussed in Section 5, there are some clear paths which must be explored for improving detection of AI-generated images

in the future. Firstly, to achieve the goal of developing a fully agnostic detector, the field must shift its focus from training on individual generators to features that apply universally to all architectures, based on the results presented in this review. While semantic features are easy targets for shortcut learning, low-level local features, including high-frequency signals and pixel correlation, appear to generalize much better. Secondly, current training methodologies must undergo a substantial overhaul. Artifact-preserving pre-processing, including cropping over resizing, as well as data augmentation techniques, along with patch-wise learning of local features, all show positive signs of improving generalization performance, as indicated by the findings reported by Li et al. and Zheng et al. [9],[12] Finally, test-time adaptation algorithms such as Cai et al.'s ATTSD [11] prove to be a compelling area to explore for distribution shift mitigation without having to retrain, although their computational overhead at inference time remains a concern for deployment.

Another key area for improvement lies in the reform of evaluation techniques, where current benchmark evaluations should be augmented by cross-semantic, cross-architectural, and robustness tests based on standard image transformations, like the Accuracy versus Average Precision gap discovered by Zheng et al. [9] It may help to develop "living" benchmarks that integrate new images generated by the most recent generative models, instead of static benchmarks which rapidly lose their relevance. Finally, Li et al. proved the necessity of efficient and lightweight models for making practical usage possible by

showing that a model with just 1.44M parameters [12] outperforms networks with tens of millions of parameters, which means that it's not needed to have conflicting priorities between efficiency and effectiveness. In closing, we have seen that overcoming the divide between benchmark evaluations and practical applications poses the main open problem in AI-generated image detection, necessitating concurrent improvements on all three fronts of training, evaluation, and deployment.

7. Conclusion

In order to analyze current approaches toward detection of images created by AI within the context of robustness and generalization, the analysis of the fundamental gap between benchmark performance and application was conducted in this review. The discussed approaches were categorized using a four-category taxonomy based on their generalization properties that includes artifact-based, spectrum-based, image encoder-based, and generator-aware categories. The key point here is that despite the good performance of currently existing detection algorithms on benchmark datasets, the problem of brittleness associated with learning shortcuts exists. These detectors tend to learn semantic cues specific to a particular generator dataset, failing to generalize and detect images regardless of the actual generator used. The occurrence of this problem can be explained by certain factors such as the inability to perform correctly when multiple generators are involved, lack of sensitivity to simple transformations and to recognize directly visualized narrow activations is another

factor. To tackle these shortcomings, changes in all these aspects simultaneously is required.

As generator models continue to improve their realism and accessibility aspects, robust detection is essential for preserving trust in digital media.

References

Ref #	Short Name	Full IEEE Reference	Type
[1]	Goodfellow et al. 2014	I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in Advances in Neural Information Processing Systems, vol. 27, 2014, pp. 2672-2680.	Foundational (GAN)
[2]	Ho et al. 2020	J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 6840-6851.	Foundational (Diffusion)
[3]	Rombach et al. 2022	R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 10684-10695.	Foundational (Latent Diffusion)
[4]	Karras et al. 2017 (ProGAN)	T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of GANs for improved quality, stability, and variation," arXiv preprint arXiv:1710.10196, 2017.	GAN Architecture
[5]	Karras et al. 2019 (StyleGAN)	T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 4401-4410.	GAN Architecture
[6]	Brock et al. 2018 (BigGAN)	A. Brock, J. Donahue, and K. Simonyan, "Large scale GAN training for high fidelity natural image synthesis," arXiv preprint arXiv:1809.11096, 2018.	GAN Architecture
[7]	Park et al. 2024	D. Park, H. Na, and D. Choi, "Performance comparison and visualization of AI-generated-image detection methods," IEEE Access, vol. 12, pp. 62609-62627, 2024.	Reviewed Paper
[8]	Song et al. 2023	H. Song, S. Huang, Y. Dong, and W.-W. Tu, "Robustness and generalizability of deepfake detection: A study with diffusion models," arXiv preprint arXiv:2309.02218, 2023.	Reviewed Paper
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[10]	Brodarič et al. 2024	M. Brodarič, V. Štruc, and P. Peer, "Cross-dataset deepfake detection: Evaluating the generalization capabilities of modern deepfake detectors," in 27th Computer Vision Winter Workshop (CVWW), Terme Olimia, Slovenia, Feb. 2024.	Reviewed Paper
[11]	Cai et al. 2025	Y. Cai, J. Tian, X. Fu, J. Dai, J. Han, and S. Lyu, "Adaptive test-time semantic debiasing for AI-generated image detection," in Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW), 2025, pp. 1554-1563.	Reviewed Paper
[12]	Li et al. 2025 (SAFE)	O. Li, J. Cai, Y. Hao, X. Jiang, Y. Hu, and F. Feng, "Improving synthetic image detection towards generalization: An image transformation perspective," in Proc. 31st ACM SIGKDD Conf. Knowl. Discovery Data Mining (KDD), Toronto, Canada, Aug. 2025, pp. 2405-2414.	Reviewed Paper
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[14]	Yadav et al. 2025	U. Yadav, P. Dasarwar, S. Bondre, and S. Kalamkar, "A hybrid approach for robust deep fake image detection using spatial and frequency domain features," Eng. Technol. Appl. Sci. Res., vol. 15, no. 3, pp. 22786-22791, 2025.	Reviewed Paper
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[21]	Verdoliva 2020	L. Verdoliva, "Media forensics and deepfakes: An overview," IEEE J. Sel. Topics Signal Process., vol. 14, no. 5, pp. 910-932, Aug. 2020.	Survey / Background

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